

Challenges of creating 3D maps with lidar technology

- **High costs and complexity.** Collecting, processing, and storing lidar data is resource-intensive.
- **Data ownership disputes.** Companies, governments, and service providers often clash over rights to lidar data.
- **Privacy concerns.** Capturing detailed images in urban areas raises legal and ethical issues.
- **Frequent updates needed.** Geographic changes require continuous mapping, increasing costs and time.
- **Access limitations.** Political and geographic restrictions may force reliance on lower-quality alternatives.

Many companies opt for cameras and radar, which offer adequate performance at a lower cost. Vehicles that include lidar often serve as a supplementary safety feature rather than the primary sensor. Its high cost may continue to hinder its broader adoption compared to more affordable technologies.

Lidar's expense is particularly challenging for autonomous vehicles, which require accurate sensing, navigation, and mapping. Although lidar provides detailed 3D data, the high costs are often prohibitive. The affordability gap with other sensors remains significant even as lidar prices decrease.

This price gap leads many manufacturers, especially those targeting lower-cost vehicles, to reconsider lidar. Its use is generally restricted to premium vehicles and autonomous taxis, where the costs are more easily justified.

Additionally, the overall expense of lidar includes not just the sensors but also integration, calibration, and maintenance, further increasing costs. In contrast, camera-based systems are cheaper and simpler to mass-produce, making them a more viable option for mass-market vehicles. Without substantial cost reductions, lidar's role in future automotive technologies may be limited.

Challenges in the lidar supply chain

While promising for enhancing vehicle autonomy, lidar technology

faces significant supply chain challenges that hinder its widespread adoption compared to more established technologies like cameras and radar. Unlike these mature sensor technologies, lidar has not yet achieved a stable supply chain, which impacts its integration into consumer vehicles.

Due to a fragmented global supplier network, the lidar industry faces challenges in supply management, quality control, and cost containment. Unlike cameras and radar, which benefit from mature supply chains and large-scale production efficiencies, lidar lacks these advantages. As a result, companies struggle to find reliable suppliers and accurately forecast long-term production costs, making large-scale adoption of lidar technology more difficult for manufacturers.

The uncertain return on investment (ROI) is a significant deterrent. The limited adoption of lidar in current consumer vehicles means manufacturers miss out on the economies of scale that help reduce costs for more established sensors. Additionally, the time needed for technological improvements and supply chain development further delays cost reductions, leaving manufacturers hesitant to invest heavily in lidar.

This reluctance is particularly noticeable in producing lower-level autonomous vehicles, where the high cost of integrating lidar is hard to justify for the minimal gains it offers over other technologies.

Companies will likely prioritise alternative sensor technologies until lidar becomes more cost-effective and demonstrates a clearer ROI.

Regionally, the adoption of lidar also varies. In China, lidar is more commonly integrated into vehicles due to government incentives, competitive pricing, and a more developed supply chain. Some Chinese companies even market lidar as a premium feature, despite the vehicles not being fully autonomous. In contrast, in other regions, lidar companies are pivoting from automotive applications to sectors like robotics, industrial automation, and logistics, which pose fewer regulatory hurdles and offer quicker ROI.

Non-automotive markets are poised for rapid growth, leveraging lidar's robustness, reliability, and 3D resolution advantages. These industries provide a broader customer base and quicker return, allowing lidar suppliers to establish a stronger foothold by addressing specific needs such as precision mapping, surveillance, and automation. As lidar technology advances and costs decrease, it can become increasingly viable for widespread adoption across these sectors, offering further opportunities for the automotive market. However, due to the same cost consideration and competition from other sensors, high-end high-performance lidars may not be the priority.

Ongoing technological uncertainty

Lidar systems, notably the advanced frequency-modulated continuous wave (FMCW) lidar, face barriers of high costs, immature supply chains, and other technical challenges. Despite its superior signal-to-noise ratio, extended range, and minimised interference, FMCW lidars have limited use now, making